

# Advanced Clustering Analysis in Data Mining

Unsupervised Learning and Cluster Robustness

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# 1 Introduction

Clustering is one of the most fundamental and widely studied tasks in unsupervised learning. Unlike supervised learning, where labeled data guides the optimization process, clustering must rely exclusively on intrinsic data properties. This makes clustering particularly sensitive to preprocessing decisions, distance metrics, feature scaling, and algorithmic assumptions.

The primary goal of this project is to perform an in-depth experimental and analytical study of clustering techniques using the Wine dataset. Rather than relying on visual inspection or external labels, all evaluations are conducted using internal validation metrics. Special emphasis is placed on robustness, distance behavior in high-dimensional space, and automated model selection.

## 2 Theoretical Background

### 2.1 Clustering Paradigms

Clustering algorithms can be categorized into several families:

- Centroid-based methods (e.g., K-Means)
- Hierarchical methods
- Density-based methods (e.g., DBSCAN)
- Model-based methods

Each paradigm makes different structural assumptions about the data.

### 2.2 Distance Metrics

Distance metrics define similarity. Common metrics include:

#### Euclidean Distance

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

#### Manhattan Distance

$$d(x, y) = \sum_{i=1}^n |x_i - y_i|$$

## Cosine Distance

$$d(x, y) = 1 - \frac{x \cdot y}{||x|| ||y||}$$

In high-dimensional spaces, distance concentration occurs, where relative differences between distances diminish.

## 3 Dataset Description

The Wine dataset consists of 178 instances with 13 continuous features. Each instance corresponds to a chemical analysis of wines derived from different cultivars.

The dataset exhibits heterogeneous feature scales. For example, alcohol concentration and magnesium values differ significantly in magnitude compared to proline concentration. This motivates normalization prior to clustering.

No missing values were detected.

## 4 Data Preprocessing

Three dataset variants were created:

### 4.1 Raw Data

The original dataset without scaling.

### 4.2 Z-Score Standardization

$$x' = \frac{x - \mu}{\sigma}$$

Standardization ensures zero mean and unit variance.

### 4.3 Min-Max Normalization

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

This transformation constrains features to the interval  $[0,1]$ .

Feature scaling is critical for distance-based clustering, as larger-scale features dominate Euclidean distance.

## 5 Exploratory Data Analysis

### 5.1 Statistical Summary

The descriptive statistics reveal considerable variation in feature ranges. Standard deviation differences justify scaling procedures.

### 5.2 Distribution Analysis

Histograms show that some features exhibit skewness and heavy tails. Certain attributes approximate Gaussian distributions, while others display asymmetric behavior.

### 5.3 Correlation Analysis

The correlation matrix highlights strong linear relationships between specific chemical properties. High correlation indicates potential redundancy and dimensional dependency.

### 5.4 PCA for Visualization

Principal Component Analysis was applied solely for visualization. The first two principal components capture a substantial fraction of total variance, allowing partial visual separation of clusters.

## 6 Distance Behavior in High-Dimensional Space

Pairwise distances were computed using Euclidean, Manhattan, and Cosine metrics.

Empirical results demonstrate distance concentration: the ratio between maximum and minimum pairwise distances decreases relative to dimensionality. This phenomenon reduces discriminative power of distance metrics.

Cosine distance exhibited more stable relative variation, suggesting its suitability in certain high-dimensional contexts.

## 7 Impact of Feature Scaling on K-Means

K-Means minimizes within-cluster variance:

$$J = \sum_{k=1}^K \sum_{x_i \in C_k} ||x_i - \mu_k||^2$$

Experiments show that unscaled data produces inferior silhouette scores, as high-variance features dominate centroid updates. Standardized data achieved superior clustering compactness and separation.

Silhouette score is defined as:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

where  $a(i)$  is intra-cluster distance and  $b(i)$  is nearest-cluster distance.

## 8 K-Means from Scratch

A full implementation of K-Means was developed including:

- Random initialization
- Assignment step
- Centroid update
- Convergence detection

Empty clusters were handled via reinitialization.

Results matched scikit-learn’s implementation in inertia values, confirming correctness.

## 9 Hierarchical Clustering Stability

Agglomerative clustering was applied with four linkage strategies. Stability was measured using the Cophenetic Correlation Coefficient.

Ward linkage achieved the highest stability, indicating better preservation of pairwise distances.

Automatic selection ensures objectivity and avoids subjective dendrogram inspection.

## 10 Density-Based Clustering

DBSCAN identifies clusters based on density connectivity.

Parameter selection:

- $k$ -nearest neighbor distance analysis
- Percentile-based  $\varepsilon$  estimation

The method detected clusters along with noise points. Performance varied depending on scaling.

## 11 Fully Automated Clustering System

A fully automated pipeline was designed to:

1. Generate scaled dataset versions
2. Perform hyperparameter search
3. Evaluate multiple algorithms
4. Select best configuration via silhouette score

Hyperparameter ranges:

- $k \in [2, 6]$  for centroid and hierarchical methods
- Multiple percentile-based  $\varepsilon$  values for DBSCAN

The final output included algorithm type, preprocessing strategy, parameters, and evaluation score.

## 12 Bonus 1: Subspace Clustering via PCA

Dimensionality reduction from 2 to 10 components was evaluated. Silhouette scores varied moderately with dimensionality. Optimal performance occurred at intermediate dimensions, balancing noise reduction and information retention.

## 13 Bonus 2: Cluster Stability under Noise

Gaussian noise was added:

$$\sigma \in \{0.01, 0.05, 0.1\}$$

Adjusted Rand Index (ARI) measured stability:

$$ARI = \frac{RI - Expected(RI)}{max(RI) - Expected(RI)}$$

Results show gradual degradation with increasing noise, indicating moderate robustness of K-Means.

## 14 Comprehensive Discussion

Major findings:

- Scaling dramatically affects centroid-based clustering.
- Distance concentration reduces discriminative power.
- Ward linkage demonstrates superior hierarchical robustness.
- DBSCAN is sensitive to  $\varepsilon$ .
- Automated pipelines improve reproducibility and objectivity.

## 15 Conclusion

This project provided a rigorous, implementation-driven analysis of clustering methods.

Key contributions include:

- Distance concentration analysis
- Scaling sensitivity study
- Custom algorithm implementation
- Stability evaluation
- Fully automated clustering framework

The results demonstrate that systematic, numerically-driven clustering analysis significantly enhances reliability and interpretability in unsupervised learning tasks.